

## Discriminant Analysis

### The DISCRIM Procedure

|                   |     |                    |     |
|-------------------|-----|--------------------|-----|
| Total Sample Size | 572 | DF Total           | 571 |
| Variables         | 8   | DF Within Classes  | 569 |
| Classes           | 3   | DF Between Classes | 2   |

|                             |     |
|-----------------------------|-----|
| Number of Observations Read | 572 |
| Number of Observations Used | 572 |

| Class Level Information |               |           |          |            |                   |
|-------------------------|---------------|-----------|----------|------------|-------------------|
| region                  | Variable Name | Frequency | Weight   | Proportion | Prior Probability |
| 1                       | 1             | 323       | 323.0000 | 0.564685   | 0.333333          |
| 2                       | 2             | 98        | 98.0000  | 0.171329   | 0.333333          |
| 3                       | 3             | 151       | 151.0000 | 0.263986   | 0.333333          |

| Within Covariance Matrix Information |                        |                                                         |
|--------------------------------------|------------------------|---------------------------------------------------------|
| region                               | Covariance Matrix Rank | Natural Log of the Determinant of the Covariance Matrix |
| 1                                    | 8                      | 51.57703                                                |
| 2                                    | 8                      | 40.30530                                                |
| 3                                    | 8                      | 47.66458                                                |
| Pooled                               | 8                      | 52.43120                                                |

## Discriminant Analysis

### The DISCRIM Procedure Test of Homogeneity of Within Covariance Matrices

| Chi-Square  | DF | Pr > ChiSq |
|-------------|----|------------|
| 2110.747587 | 72 | <.0001     |

Since the Chi-Square value is significant at the 0.1 level, the within covariance matrices will be used in the discriminant function.  
Reference: Morrison, D.F. (1976) Multivariate Statistical Methods p252.

## Discriminant Analysis

### The DISCRIM Procedure

| Generalized Squared Distance to region |          |          |          |
|----------------------------------------|----------|----------|----------|
| From region                            | 1        | 2        | 3        |
| 1                                      | 51.57703 | 1269     | 1591     |
| 2                                      | 89.09331 | 40.30530 | 81.29521 |
| 3                                      | 76.08996 | 96.69547 | 47.66458 |

## Discriminant Analysis

### The DISCRIM Procedure Classification Summary for Calibration Data: WORK.OIL Resubstitution Summary using Quadratic Discriminant Function

| Number of Observations and Percent Classified into region |
|-----------------------------------------------------------|
|-----------------------------------------------------------|

| From region | 1             | 2            | 3             | Total         |
|-------------|---------------|--------------|---------------|---------------|
| 1           | 323<br>100.00 | 0<br>0.00    | 0<br>0.00     | 323<br>100.00 |
| 2           | 0<br>0.00     | 98<br>100.00 | 0<br>0.00     | 98<br>100.00  |
| 3           | 0<br>0.00     | 0<br>0.00    | 151<br>100.00 | 151<br>100.00 |
| Total       | 323<br>56.47  | 98<br>17.13  | 151<br>26.40  | 572<br>100.00 |
| Priors      | 0.33333       | 0.33333      | 0.33333       |               |

| Error Count Estimates for region |        |        |        |        |
|----------------------------------|--------|--------|--------|--------|
|                                  | 1      | 2      | 3      | Total  |
| Rate                             | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Priors                           | 0.3333 | 0.3333 | 0.3333 |        |

## Discriminant Analysis

The DISCRIM Procedure  
 Classification Summary for Calibration Data: WORK.OIL  
 Cross-validation Summary using Quadratic Discriminant Function

| Number of Observations and Percent Classified into region |               |              |               |               |
|-----------------------------------------------------------|---------------|--------------|---------------|---------------|
| From region                                               | 1             | 2            | 3             | Total         |
| 1                                                         | 323<br>100.00 | 0<br>0.00    | 0<br>0.00     | 323<br>100.00 |
| 2                                                         | 0<br>0.00     | 98<br>100.00 | 0<br>0.00     | 98<br>100.00  |
| 3                                                         | 0<br>0.00     | 0<br>0.00    | 151<br>100.00 | 151<br>100.00 |
| Total                                                     | 323<br>56.47  | 98<br>17.13  | 151<br>26.40  | 572<br>100.00 |
| Priors                                                    | 0.33333       | 0.33333      | 0.33333       |               |

| Error Count Estimates for region |        |        |        |        |
|----------------------------------|--------|--------|--------|--------|
|                                  | 1      | 2      | 3      | Total  |
| Rate                             | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Priors                           | 0.3333 | 0.3333 | 0.3333 |        |

## Discriminant analysis — nine-area classification

The DISCRIM Procedure

|                   |     |                    |     |
|-------------------|-----|--------------------|-----|
| Total Sample Size | 572 | DF Total           | 571 |
| Variables         | 8   | DF Within Classes  | 563 |
| Classes           | 9   | DF Between Classes | 8   |

|                             |     |
|-----------------------------|-----|
| Number of Observations Read | 572 |
| Number of Observations Used | 572 |

Class Level Information

| area            | Variable Name   | Frequency | Weight   | Proportion | Prior Probability |
|-----------------|-----------------|-----------|----------|------------|-------------------|
| Calabria        | Calabria        | 56        | 56.0000  | 0.097902   | 0.111111          |
| Coast-Sardinia  | Coast-Sardinia  | 33        | 33.0000  | 0.057692   | 0.111111          |
| East-Liguria    | East-Liguria    | 50        | 50.0000  | 0.087413   | 0.111111          |
| Inland-Sardinia | Inland-Sardinia | 65        | 65.0000  | 0.113636   | 0.111111          |
| North-Apulia    | North-Apulia    | 25        | 25.0000  | 0.043706   | 0.111111          |
| Sicily          | Sicily          | 36        | 36.0000  | 0.062937   | 0.111111          |
| South-Apulia    | South-Apulia    | 206       | 206.0000 | 0.360140   | 0.111111          |
| Umbria          | Umbria          | 51        | 51.0000  | 0.089161   | 0.111111          |
| West-Liguria    | West-Liguria    | 50        | 50.0000  | 0.087413   | 0.111111          |

| Within Covariance Matrix Information |                        |                                                         |
|--------------------------------------|------------------------|---------------------------------------------------------|
| area                                 | Covariance Matrix Rank | Natural Log of the Determinant of the Covariance Matrix |
| Calabria                             | 8                      | 45.53284                                                |
| Coast-Sardin                         | 8                      | 34.57630                                                |
| East-Liguria                         | 8                      | 42.87014                                                |
| Inland-Sardi                         | 8                      | 37.97195                                                |
| North-Apulia                         | 8                      | 45.19879                                                |
| Sicily                               | 8                      | 48.68371                                                |
| South-Apulia                         | 8                      | 48.35201                                                |
| Umbria                               | 8                      | 31.98920                                                |
| West-Liguria                         | 8                      | 43.43016                                                |
| Pooled                               | 8                      | 48.77624                                                |

### Discriminant analysis — nine-area classification

The DISCRIM Procedure  
Test of Homogeneity of Within Covariance Matrices

| Chi-Square  | DF  | Pr > ChiSq |
|-------------|-----|------------|
| 2689.722277 | 288 | <.0001     |

Since the Chi-Square value is significant at the 0.1 level, the within covariance matrices will be used in the discriminant function.  
Reference: Morrison, D.F. (1976) Multivariate Statistical Methods p252.

### Discriminant analysis — nine-area classification

The DISCRIM Procedure

| Generalized Squared Distance to area |           |                |              |                 |              |           |              |           |              |
|--------------------------------------|-----------|----------------|--------------|-----------------|--------------|-----------|--------------|-----------|--------------|
| From area                            | Calabria  | Coast-Sardinia | East-Liguria | Inland-Sardinia | North-Apulia | Sicily    | South-Apulia | Umbria    | West-Liguria |
| Calabria                             | 45.53284  | 1986           | 2065         | 1534            | 102.06921    | 63.03648  | 71.16537     | 3930      | 1977         |
| Coast-Sardinia                       | 204.81354 | 34.57630       | 197.34800    | 110.01413       | 253.37632    | 171.05501 | 93.34046     | 4356      | 219.09214    |
| East-Liguria                         | 111.43362 | 477.26843      | 42.87014     | 263.78791       | 99.82658     | 108.54161 | 102.88254    | 216.77091 | 116.78191    |
| Inland-Sardinia                      | 163.93381 | 83.49209       | 125.35352    | 37.97195        | 147.79400    | 120.85087 | 85.96622     | 2195      | 188.63474    |
| North-Apulia                         | 108.09092 | 3183           | 2942         | 2289            | 45.19879     | 55.87050  | 83.70884     | 2515      | 2817         |
| Sicily                               | 59.43233  | 3577           | 3725         | 2760            | 86.94219     | 48.68371  | 69.56056     | 4892      | 3505         |
| South-Apulia                         | 94.80073  | 1419           | 2271         | 1138            | 167.51953    | 82.72006  | 48.35201     | 8063      | 1500         |

| Generalized Squared Distance to area |           |                |              |                 |              |           |              |           |              |
|--------------------------------------|-----------|----------------|--------------|-----------------|--------------|-----------|--------------|-----------|--------------|
| From area                            | Calabria  | Coast-Sardinia | East-Liguria | Inland-Sardinia | North-Apulia | Sicily    | South-Apulia | Umbria    | West-Liguria |
| Umbria                               | 98.87493  | 550.31284      | 57.26577     | 401.97420       | 114.44790    | 106.69007 | 105.29592    | 31.98920  | 118.88048    |
| West-Liguria                         | 172.49356 | 232.48480      | 84.02640     | 190.98996       | 150.53452    | 271.75506 | 105.67240    | 922.86643 | 43.43016     |

### Discriminant analysis — nine-area classification

The DISCRIM Procedure  
 Classification Summary for Calibration Data: WORK.OIL  
 Resubstitution Summary using Quadratic Discriminant Function

| Number of Observations and Percent Classified into area |             |                |              |                 |              |              |              |              |              |               |
|---------------------------------------------------------|-------------|----------------|--------------|-----------------|--------------|--------------|--------------|--------------|--------------|---------------|
| From area                                               | Calabria    | Coast-Sardinia | East-Liguria | Inland-Sardinia | North-Apulia | Sicily       | South-Apulia | Umbria       | West-Liguria | Total         |
| Calabria                                                | 54<br>96.43 | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 1<br>1.79    | 1<br>1.79    | 0<br>0.00    | 0<br>0.00    | 56<br>100.00  |
| Coast-Sardinia                                          | 0<br>0.00   | 33<br>100.00   | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 33<br>100.00  |
| East-Liguria                                            | 0<br>0.00   | 0<br>0.00      | 50<br>100.00 | 0<br>0.00       | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 50<br>100.00  |
| Inland-Sardinia                                         | 0<br>0.00   | 0<br>0.00      | 0<br>0.00    | 65<br>100.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 65<br>100.00  |
| North-Apulia                                            | 0<br>0.00   | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 25<br>100.00 | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 25<br>100.00  |
| Sicily                                                  | 0<br>0.00   | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 36<br>100.00 | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 36<br>100.00  |
| South-Apulia                                            | 1<br>0.49   | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 5<br>2.43    | 200<br>97.09 | 0<br>0.00    | 0<br>0.00    | 206<br>100.00 |
| Umbria                                                  | 0<br>0.00   | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 51<br>100.00 | 0<br>0.00    | 51<br>100.00  |
| West-Liguria                                            | 0<br>0.00   | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 50<br>100.00 | 50<br>100.00  |
| Total                                                   | 55<br>9.62  | 33<br>5.77     | 50<br>8.74   | 65<br>11.36     | 25<br>4.37   | 42<br>7.34   | 201<br>35.14 | 51<br>8.92   | 50<br>8.74   | 572<br>100.00 |
| Priors                                                  | 0.11111     | 0.11111        | 0.11111      | 0.11111         | 0.11111      | 0.11111      | 0.11111      | 0.11111      | 0.11111      |               |

| Error Count Estimates for area |          |                |              |                 |              |        |              |        |              |        |
|--------------------------------|----------|----------------|--------------|-----------------|--------------|--------|--------------|--------|--------------|--------|
|                                | Calabria | Coast-Sardinia | East-Liguria | Inland-Sardinia | North-Apulia | Sicily | South-Apulia | Umbria | West-Liguria | Total  |
| Rate                           | 0.0357   | 0.0000         | 0.0000       | 0.0000          | 0.0000       | 0.0000 | 0.0291       | 0.0000 | 0.0000       | 0.0072 |
| Priors                         | 0.1111   | 0.1111         | 0.1111       | 0.1111          | 0.1111       | 0.1111 | 0.1111       | 0.1111 | 0.1111       |        |

### Discriminant analysis — nine-area classification

The DISCRIM Procedure  
 Classification Summary for Calibration Data: WORK.OIL  
 Cross-validation Summary using Quadratic Discriminant Function

| Number of Observations and Percent Classified into area |             |                |              |                 |              |           |              |           |              |              |
|---------------------------------------------------------|-------------|----------------|--------------|-----------------|--------------|-----------|--------------|-----------|--------------|--------------|
| From area                                               | Calabria    | Coast-Sardinia | East-Liguria | Inland-Sardinia | North-Apulia | Sicily    | South-Apulia | Umbria    | West-Liguria | Total        |
| Calabria                                                | 53<br>94.64 | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 2<br>3.57 | 1<br>1.79    | 0<br>0.00 | 0<br>0.00    | 56<br>100.00 |
| Coast-Sardinia                                          | 0<br>0.00   | 32<br>96.97    | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 0<br>0.00 | 1<br>3.03    | 0<br>0.00 | 0<br>0.00    | 33<br>100.00 |

| Number of Observations and Percent Classified into area |            |                |              |                 |              |             |              |              |              |               |
|---------------------------------------------------------|------------|----------------|--------------|-----------------|--------------|-------------|--------------|--------------|--------------|---------------|
| From area                                               | Calabria   | Coast-Sardinia | East-Liguria | Inland-Sardinia | North-Apulia | Sicily      | South-Apulia | Umbria       | West-Liguria | Total         |
| East-Liguria                                            | 0<br>0.00  | 0<br>0.00      | 49<br>98.00  | 0<br>0.00       | 0<br>0.00    | 0<br>0.00   | 0<br>0.00    | 0<br>0.00    | 1<br>2.00    | 50<br>100.00  |
| Inland-Sardinia                                         | 0<br>0.00  | 1<br>1.54      | 0<br>0.00    | 64<br>98.46     | 0<br>0.00    | 0<br>0.00   | 0<br>0.00    | 0<br>0.00    | 0<br>0.00    | 65<br>100.00  |
| North-Apulia                                            | 0<br>0.00  | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 23<br>92.00  | 1<br>4.00   | 1<br>4.00    | 0<br>0.00    | 0<br>0.00    | 25<br>100.00  |
| Sicily                                                  | 0<br>0.00  | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 34<br>94.44 | 2<br>5.56    | 0<br>0.00    | 0<br>0.00    | 36<br>100.00  |
| South-Apulia                                            | 1<br>0.49  | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 5<br>2.43   | 200<br>97.09 | 0<br>0.00    | 0<br>0.00    | 206<br>100.00 |
| Umbria                                                  | 0<br>0.00  | 0<br>0.00      | 0<br>0.00    | 0<br>0.00       | 0<br>0.00    | 0<br>0.00   | 0<br>0.00    | 51<br>100.00 | 0<br>0.00    | 51<br>100.00  |
| West-Liguria                                            | 0<br>0.00  | 0<br>0.00      | 1<br>2.00    | 0<br>0.00       | 0<br>0.00    | 0<br>0.00   | 0<br>0.00    | 0<br>0.00    | 49<br>98.00  | 50<br>100.00  |
| Total                                                   | 54<br>9.44 | 33<br>5.77     | 50<br>8.74   | 64<br>11.19     | 23<br>4.02   | 42<br>7.34  | 205<br>35.84 | 51<br>8.92   | 50<br>8.74   | 572<br>100.00 |
| Priors                                                  | 0.11111    | 0.11111        | 0.11111      | 0.11111         | 0.11111      | 0.11111     | 0.11111      | 0.11111      | 0.11111      |               |

| Error Count Estimates for area |          |                |              |                 |              |        |              |        |              |        |
|--------------------------------|----------|----------------|--------------|-----------------|--------------|--------|--------------|--------|--------------|--------|
|                                | Calabria | Coast-Sardinia | East-Liguria | Inland-Sardinia | North-Apulia | Sicily | South-Apulia | Umbria | West-Liguria | Total  |
| Rate                           | 0.0536   | 0.0303         | 0.0200       | 0.0154          | 0.0800       | 0.0556 | 0.0291       | 0.0000 | 0.0200       | 0.0338 |
| Priors                         | 0.1111   | 0.1111         | 0.1111       | 0.1111          | 0.1111       | 0.1111 | 0.1111       | 0.1111 | 0.1111       |        |